
The PETMAN and Atlas Robots at Boston Dynamics

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Abstract

In 2008, Boston Dynamics began the development of a series of humanoid robots. The goal was to build bipeds with advanced mobility and dexterity. Our first bipeds, the PETMAN robots, later evolved into three different versions of Atlas. Collectively we programmed these robots to walk at human speeds, cross rock fields, climb stairs and inclines, and negotiate outdoor terrain. We also experimented with augmenting balance using angular momentum. Here we recount the development of these machines and the locomotion behaviors we were able to demonstrate with them.

Keywords

Boston dynamics · PETMAN · Atlas · Bipeds · DRC · Balance · Angular momentum

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1 Introduction

In December of 2008 Boston Dynamics [4] began the development of PETMAN, a humanoid robot meant to test chemical protective clothing for the Army [1]. PETMAN was the first of a series of hydraulic humanoids we developed as we explored how to build robots with advanced mobility and dexterity. We built two prototypical bipeds from BigDog leg parts, the PETMAN robot, and three different versions of Atlas which evolved from the PETMAN design. Collectively we programmed these robots to walk at typical human walking speeds up to 2 m/s (4.5 mph), cross rock fields reliably, walk up 28 degree inclines, and walk over 1 mile in outdoor terrain without falling. We also experimented with using the upper body of these robots to augment balance during locomotion. We developed stair and step climbing strategies in which the robot leaned on outstretched arms to maintain balance while climbing. One robot ascended a 50 degree set of stairs at twice the speed of contemporaneous bipedal robots. We made significant progress in using upper body momentum to maintain balance while walking and standing on a single foot. In the following we recount the development of these machines and the locomotion behaviors we were able to demonstrate with them. Table 1 gives an outline of the history of Boston Dynamics’ bipedal robots to date. The table shows each robot and when it was active, as well as key goals and behaviors that the robot achieved.

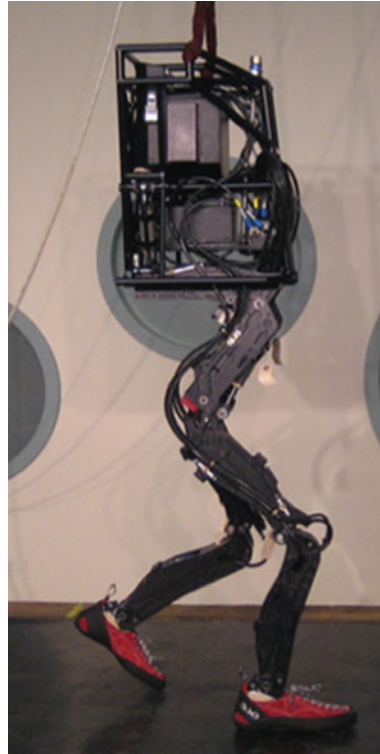
Table 1 A history of Boston Dynamics’ bipedal robots through 2015 [6]

Biped Robot	PETProto	PETMAN	AtlasProto	Atlas	Atlas-DRC	Atlas-Unplugged
Picture & Year	 2009	 2011	 2012	 2013	 2013	 2015
Key Behaviors	Quickly demonstrating fast, smooth human-like walking.	Anthropometric robot for testing chemical protective clothing; Demonstrating human-like walking & calisthenics.	Dynamically negotiating stairs & indoor obstacles; Assisting balance by bracing with arms.	Indoor & outdoor rough terrain & trails; Fine balance using angular momentum.	DARPA DRC trials; Perception, mobility, & manipulation in simulated disaster scenario.	DARPA DRC final competition; Perception, mobility, & manipulation in simulated disaster scenario.

2 The PETMAN Robots

The purpose of PETMAN was to serve as a robotic human-surrogate for testing chemical protective clothing. The robot was designed to wear protective clothing in an environmentally controlled test chamber, where it is exposed to chemical agents as it walks and does basic calisthenics. Chemical sensors embedded in the skin of the robot measure if, when, and where chemical agents are detected within the suit. The robot performs these tests under controlled temperature and wind conditions. A treadmill and turntable integrated into the wind tunnel chamber allow for sustained walking experiments that can be variably oriented relative to the wind. One of the key requirements for the PETMAN robot was being able to perform human-like motions, especially a natural, human-like walking gait, something which had not been demonstrated in biped robots to date. We knew the development of a new walking gait could not wait for the completed PETMAN robot, which would be over 2 years in the making. In order to make walking progress early in the program, we quickly developed a simple biped called the PETMAN Prototype (or PETProto) (Figs. 1 and 2).

Fig. 1 PETProto, Boston Dynamics' first bipedal robot



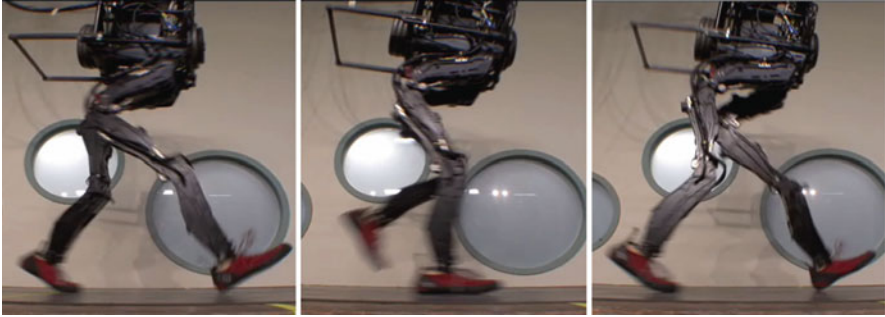


Fig. 2 PETProto walked with a natural heel-to-toe gait at speeds up to 2 m/s (4.5 mph). The robot was built quickly using BigDog hardware in order to speed up the development of software for PETMAN

3 PETProto

PETProto was the first bipedal robot built at Boston Dynamics. We built PETProto in about 4 months by adapting the BigDog [3] leg design and electronics. PETProto had two, five degree-of-freedom (DoF) legs: the uppermost DoF being abduction/adduction at the hip, and remaining four DoF nominally moving the leg parallel to the sagittal plane. PETProto was about 1.5 m tall, with the hips about 1 m above the ground. The hips had a 19 cm lateral separation, which is similar to that of a human.

Mass (kg)	44
Overall height (cm)	150
#Actuated DoFs (Total: Legs, Back, Arms, Neck)	10: 2×5, 0, 0, 0

With PETProto, we were able to demonstrate a dynamic, high-speed walking gait. The robot, starting from a stand, was able to walk at varying speeds as high as 2 m/s (4.5 mph). Qualitatively the gait demonstrated an extended leg posture during heel strike and included a smooth, human-like heel-to-toe transition during stance. The robot walking was robust to moderate disturbances allowing it maintain position on a treadmill as the treadmill speed was increased or decreased.

The algorithm used for this walking introduced a new method (for Boston Dynamics) of maintaining balance. That algorithm was based on regulating balance via the *timing* of foot placement, as opposed to the location of foot placement. This helped significantly in making the walking look smooth and natural.

The approximately 44 kg robot was able to carry a payload of 40 kg. While walking, it could regain its balance when pushed moderately from the side, though disturbances that caused large crossover steps between the legs often caused a

Table 2 Various foot designs used with PETMAN and the Atlas robots

				
PETMAN		Atlas DRC		Atlas “Duckling”
Foot Style	Length x Width (mm x mm)	Weight (kg)	Tread	Sensing
PETMAN	238 x 65	0.5	Human footwear over foam-rubber faux-foot	None
Atlas DRC	256 x 121	2	Custom replaceable Vibram® soles	Z force & CoP via custom strain-gauge load-cells
Atlas “Duckling”	197 x 95	1.4	Custom replaceable Vibram® soles	Z force & CoP via custom strain-gauge load-cells

fall. PETProto was also able to walk, without control modifications, with varying footwear (from slender climbing shoes – size 10 US, with a maximum width of 9.5 cm – to large army boots), and with a hydraulic tether attached to one leg just above the ankle (to simulate how PETMAN itself would eventually be tethered to hydraulic power).

It is noteworthy that the lack of hip and ankle joints for internal/external rotation did not appear to hinder robust walking. One limitation of PETProto was its relative inability to perform a robust balanced stand. The robot could not easily return to a standing posture once it was walking, something PETMAN would need to demonstrate. Both PETProto and PETMAN used a compliant, carbon-fiber prosthetic foot (see Table 2) that could fit inside unaltered human footwear, a requirement for the robot. This arrangement made it difficult to sense or control the center-of-pressure (CoP) on the sole of the shoe, which also made controlling standing postures difficult.

PETProto walking experiments were helpful in the development of PETMAN by providing empirical data for actuator and sensing requirements, as well as a testbed for the development of algorithms that could regulate balance during a dynamic walking gait.

4 PETMAN

The proposed mission for PETMAN drove two key requirements of the development program for this anthropometric robot: (1) it should be the size and shape of a 50th percentile human male and (2) it should be able to perform human-like motions, including a natural walking gait. In contrast to the relative simplicity of

PETProto, PETMAN ultimately needed to fit 29 actuated degrees of freedom into the envelope of an average human male, and the resulting system needed to produce the large ranges of motion and strength required for natural human-like behavior (Figs. 3 and 4).

Mass (kg)	108
Overall height (cm)	170
#Actuated DoFs (Total: Legs, Back, Arms, Neck)	29: 2×6, 3, 2×6 ^a , 2

^aw/o hands

The resulting PETMAN robot was a freestanding biped powered by an off-board hydraulic power unit (HPU) via a tether that entered the right ankle. Onboard systems include actuation, thermal management, sensing, computation, and controls.

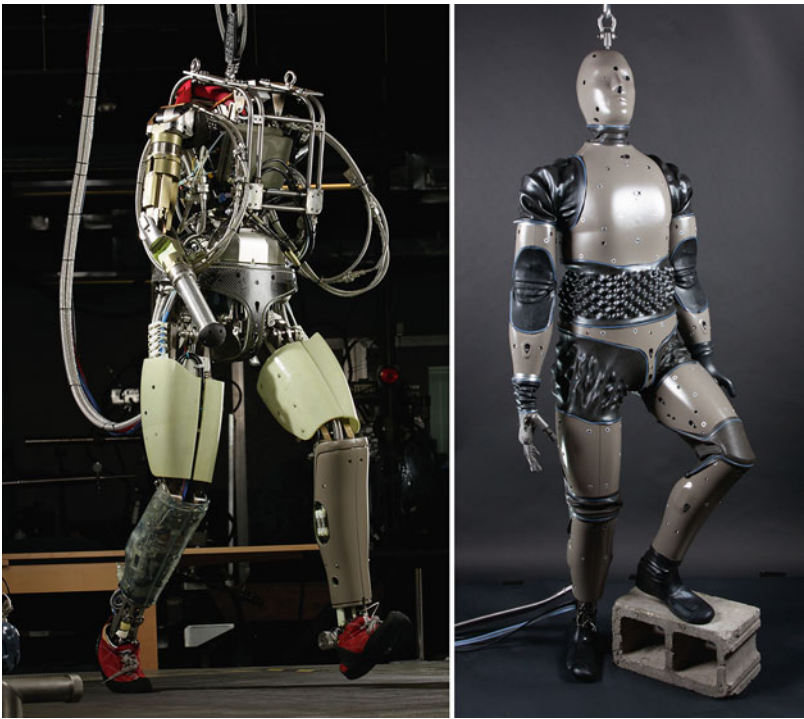


Fig. 3 The Individual Protective Equipment Mannequin System will be used to test chemical protective clothing. It is comprised of the PETMAN robot, the Intelligent Safety Harness, a containment chamber and wind tunnel, treadmill and turntable, and a control station for preparing and monitoring tests. A preliminary version of PETMAN is shown on the *left*, the final version with anthropometrically correct form, sealed skins and hydraulic tether attached at the ankle is shown on the *right*

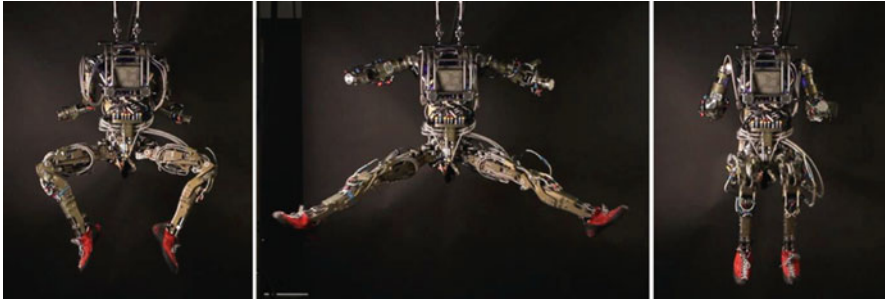


Fig. 4 PETMAN was designed to achieve natural human-like motion with large ranges of motion and high strength

The robot contained over 90 integrated sensors. Inertial sensors measure the attitude and acceleration of the body, while joint sensors measure the motion and force of the actuators. The onboard computer integrated information from these sensors to provide estimates of how PETMAN was moving in space. Other off-board sensors in the HPU monitored PETMAN's homeostasis: hydraulic pressure, flow and temperature. Hydraulic flow to the actuators was regulated by aerospace-quality servo-valves. The robot had the following actuated joints (joint: #DoF): ankle:2, knee:1, hip:3, back:3, shoulder:3, elbow:1, wrist:2, neck:2. Each wrist had a passive twist DoF. The hands featured articulated fingers actuated by a single hydraulic valve allowing the hand to open or close. The robot weighed about 108 kg with an additional payload capacity of 23 kg (so it could wear body armor during tests) and was about 170 cm tall.

The robot skin was comprised of hard shells that incorporated thermal regulation and perspiration. Included in the power tether were cooled water (for regulating skin temperature and delivering simulated sweat), electrical power, and communications to and from the robot. The shells were connected by flexible joint sleeves. The shells and sleeves created an airtight seal that nominally prevented any chemicals from passing through the skin of the robot. Nevertheless, the robot was designed to tolerate chemical intrusion and decontamination if necessary.

An onboard computer read the sensors, controlled PETMAN's behavior, and handled communications with a remote human operator. It also performed both low-level and high-level control functions. Onboard communication used a modified CANBus protocol to support a 1 kHz control loop. The control computer also recorded large amounts of engineering data for performance and failure analysis, and operational support.

PETMAN demonstrated dynamically balanced, heel-to-toe walking at a range of speeds up to and including 1.4 m/s (3 mph). Using motion scripting software we created, the robot demonstrated 2 h endurance tests in which it performed programmed sequences of motions including stand, squat, squat while turning, side-step with arms raised overhead, walk, and high-stepping walk. The control architecture allowed for goal data (from motion capture of human walking) to

influence the motion of the robot, especially with the upper body or where kinematic redundancy was present. Using such goal data during walking, PETMAN could mimic human upper-body motion in the back, shoulders, and arms. This upper-body motion also played an important control role in counteracting inertial reactions due to leg swing accelerations. This made the overall, whole-body motion during the gait look more natural. Also, like PETProto, PETMAN was able to walk with different types of footwear and with the hydraulic tether attached just above the right ankle.

5 The Atlas Robots

The Atlas robots were developed to explore new regimes of mobility, including how upper body limbs may be used to enhance rough terrain capability. In some cases, where balance was especially challenging, we were interested in exploring how to “cheat” by using arms to push against the environment (i.e., “bracing”), and thus help stabilize the robot. This investigation began with AtlasProto, a quick modification to PETProto and then progressed to a new robot design we called Atlas. With Atlas, we studied both reactive and fine balance. The Atlas robots were also developed to support the DARPA Robotics Challenge, an international contest designed to foster the development of advanced robotics technology. This work is discussed below (Fig. 5).

6 AtlasProto

AtlasProto was created by simply adding arms to PETProto. Each arm had a 2 DoF shoulder, a 1 DoF elbow, a passive spring-loaded prismatic joint in the forearm, and a simple hemispherical “hand.” The springs and sensing in this forearm joint allowed AtlasProto to sense when the hand was making contact with the environment, and how much force was being produced axially along the forearm.

Mass (kg)	61
Overall height (cm)	150
#Actuated DoFs (Total: Legs, Back, Arms, Neck)	16: 2×5, 0, 2×3, 0

With AtlasProto, we tackled two new tasks for our biped work: dynamically climbing stairs and negotiating a simulated disaster hallway with large discrete obstacles (a.k.a. the “obstructed-hallway”). Both of these tasks used bracing with the arms to help the robot balance. In the case of the dynamic stair-climb, the robot braced against a single wall on one side of the stairs, while in the case of the obstructed-hallway, the robot braced against both walls that enclosed the hallway (Fig. 6)

Fig. 5 To build AtlasProto we added a simplified variant of PETMAN arms to the PETProto robot.

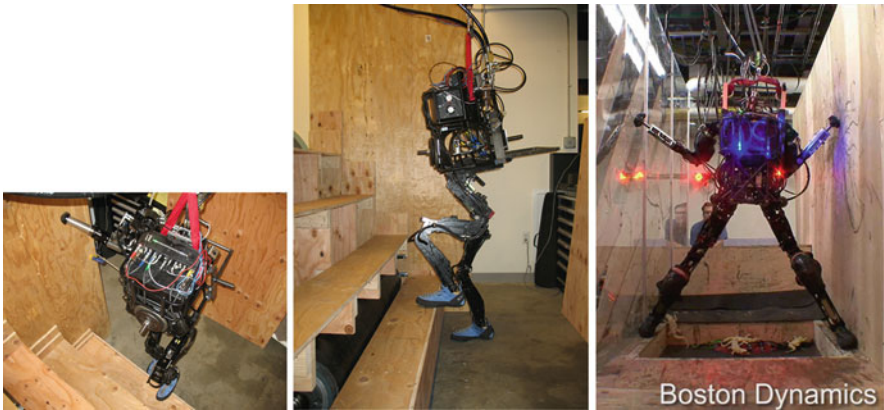
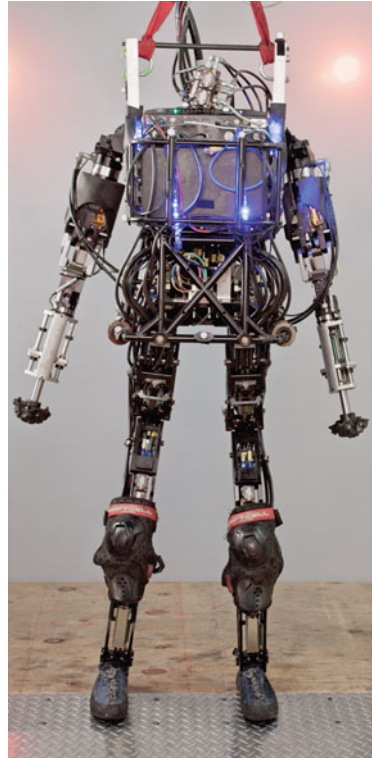


Fig. 6 AtlasProto climbing a 50 degree set of stairs and crossing the “snake-pit”

One key part of the dynamic stair-climb was the need to use foot-placement timing to achieve balance in the sagittal plane. This reintroduced the strategy used with PETProto walking, and was necessary because (i) in order to step once on each stair, the robot had to place feet at specific foot-holds in the world (in contrast to

placing feet relative to the body), and (ii) during climbing, interference between the robot's knee and the upcoming stair meant that the robot would need to climb using only the forefoot (i.e., the portion of the sole between the ball of the foot and the toes) and a relatively narrow strip of the stair tread nearest the edge of the step. Overall, this was a task very much like normal human stair climbing!

Whereas the dynamic stair climbing used only one of AtlasProto's arms, the obstructed hallway used both arms. The standard 48 in. hallway width worked well with AtlasProto, and allowed the robot to brace against both walls to help in negotiating the obstructions in its path. These obstructions were two: a 51 cm high by 30 cm wide discrete block (the "big-step"), and a 1 m $\times$ 1 m "snake-pit" where the floor of the hallway was missing. Negotiating these two obstacles required developing three discrete "you-only-have-one-shot" behaviors: getting up on top of the big-step, jumping down from the big-step, and getting across the snake-pit.

These AtlasProto behaviors were nearly diametrical to our previous work with continuous dynamic walking, wherein we only concerned ourselves with, for instance, foot placement relative to the body. In other words, previously the key control input for regulating balance did not concern itself with where the body was in the world, and assumed ground was available, relative to the body, for foot placement. With these new tasks, the robot would have to negotiate past fixed obstacles in the world and would therefore need to monitor (and control) where it was relative to the obstacle, and also how it was placing its feet or hands relative to the obstacle. For this reason, a Vicon motion-capture system was used to sense the location of the robot, providing real-time feedback to the control system. The robot body included retro-reflective markers and the terrain was constructed to make the robot body easily visible to the Vicon cameras (in the case of the hallway, one wall was constructed of transparent Lexan). With AtlasProto, we achieved a success rate of about two-thirds with these difficult tasks but also gained some enlightening experience in moving dynamically and balancing in very constrained terrain.

7 Atlas

Atlas combined the lower body of PETMAN with a simplified torso, articulated shoulders, and club-like arms. The arms were active halteres (cf. insect halteres) and were contributors to angular momentum control. Perhaps the main control-enabling hardware change that Atlas introduced was CoP sensing on the feet, and the robot went through a series of foot designs, three of which are shown in Table 2. The light-weight, sensor-less PETMAN feet were replaced with larger, flat, CoP-sensing Atlas-DRC feet, which were in turn carried over to the DRC robots. This design was then refined into a smaller instrumented foot (named "Duckling") for Atlas fieldwork. The addition of these stiffer, instrumented feet, and the ability to do active closed-loop CoP control, gave Atlas much better stability during standing than PETMAN.

Mass (kg)	95
Overall height (cm)	165
#Actuated DoFs (Total: Legs, Back, Arms, Neck)	19: 2×6, 3, 2×2, 0

With Atlas, we initially developed behavior in two directions: walking over rough and inclined terrain using reactive foot-placement control, and developing our first fine balance control using improved state estimation and angular momentum. Later, we added a third direction by extending the balance algorithms developed on PETMAN/AtlasProto to work in multiple dimensions. This more directly used the concept of the “Capture Point” – a theoretical location on the ground where placing the CoP of a linear (constant CoM height) inverted pendulum (LIP) would cause the LIP to have zero orbital energy [2, 5]. The Capture Point allows the walking controller to relate the dynamics of the center-of-mass (CoM) in both the lateral and sagittal directions to the contact forces on the ground, i.e., where the feet are placed. This became a more general approach to locomotion tasks where precision foot placement in the world is important, such as stair climbing, obstacle avoidance, or approaching an object to manipulate.

Using reactive foot placement for balance, Atlas was able to traverse a fairly wide array of terrain. These included rock fields, hills and grassy slopes, curbs, and 1 mile of wooded trails. It could also tolerate unexpected steps up of 22 cm. When outdoors, the robot was powered, via tether, from a portable HPU that was carried in a separate chase vehicle. The robot was also equipped with crash protection for the inevitable spills that are part of controls development (Fig. 7).

As part of its rough terrain control, Atlas had algorithms that allowed the robot to react to challenging balance situations by stepping more quickly – thus increasing the effective “bandwidth” of the foot-placement control. We also began to give the robot perception of the terrain directly in front of it by using vision-based



Fig. 7 The Atlas robot was configured for rough terrain mobility development by adding foam crash padding to the arms and torso and replacing the distal arm joints with simple weights. A downward facing stereo camera was added to the head. Portable hydraulic and electric power units were mounted to a small vehicle fit with a boom and winch for transporting the robot between experiments

information from a stereo camera pair. Much of this work paralleled our previous history with Boston Dynamics quadruped robots, BigDog and LS3. This includes adding ground-plane estimation – a fusion of gyro and kinematic data to estimate ground inclination below the robot – and the appropriate use of that information in negotiating hills.

A new research area that we wanted to begin to tackle with Atlas was developing and demonstrating much finer balance control than our previous robots. The impetus for this was our experiences with the AtlasProto tasks, wherein we needed to control falling direction with limited CoP control and no foot-placement input. Specifically, we wanted to be able to control how the robot was moving without taking **any** steps, and we began by investigating how well we could get the robot to balance while standing on one leg. At low velocities, the perfectly upright position of an inverted pendulum is an unstable equilibrium position (or more correctly, a saddle-point in state space). In the vicinity of this location, normally small control actions, or small model errors, can become dominant since they can move the pendulum across equilibrium and thus reverse the falling direction. We sought to be able to control balance around this unstable equilibrium by using a combination of CoP and angular momentum control strategies.

Towards this end, we developed modeling and control for Atlas that enabled the robot to estimate, in real time, both where its effective CoM was, relative to this unstable equilibrium, and the CoM velocity. The controller would then take appropriate actions to stabilize this effective CoM over the center of the foot. By monitoring the evolution of linear and angular momentum, we began to give Atlas an intrinsic sense of balance, allowing the control system to gain a measure of “how balanced am I?” – a set of information we began to call the robot’s “balance state.” This was done in the presence of modeling errors in both effective CoM location (typically only known to within $\pm 2\text{--}3$ cm, which included unsensed flex and backlash in the limbs of the robot), and unknown disturbance forces from the power tether and wrangle. Using these control techniques, we could disturb Atlas, while standing on one leg, by swinging a 20 lb (9 kg) medicine ball into it from the side (Fig. 8). While rejecting this disturbance and remaining standing on the same leg, the robot could typically outperform an unsighted human subject attempting the same task. We also demonstrated a “balance-beam” performance with Atlas. Starting from standing on two legs, the robot could shift its weight and stand with just one leg on the narrow edge of a $2'' \times 4''$ piece of lumber (Fig. 8). This balance-beam gave the robot only 3 cm of lateral CoP authority, which enabled us to demonstrate the control system’s emphasis on sensing balance state and maintaining balance by modulating angular momentum through body posture.

Further development followed by integrating this balance controller into walking. As mentioned above, historically our baseline walking control achieved balance via reactive foot placement. Robot motion was assumed to be that of an unrestricted linear inverted pendulum, where the ground reaction force nominally pointed through the CoM. Our experience with bipeds was showing us that this assumption was often problematic. This new balance controller altered this paradigm by giving us a better understanding of how the robot’s motion was affecting the ground

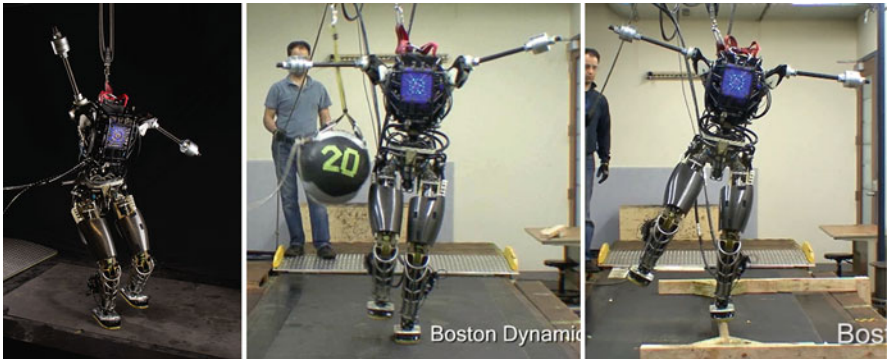


Fig. 8 The Atlas robot could coordinate changes in its angular momentum to aid in balance. It could maintain balance on a single leg while struck with a 20 lb medicine ball, and also balanced on a narrow “balance beam” (the narrow edge of a 2” × 4” piece of lumber). The robot used rotational movement of the arms, torso, and swing leg as well as movement of the CoP below the stance foot to achieve this balance control

reaction force. Thus, it allowed us to *not* point the reaction through the CoM in a coordinated way – hence the induction of angular momentum changes. Comparisons of walking with and without this attention to angular momentum showed that we could improve the overall lateral balance performance, allowing the robot to maintain a tighter centerline when disturbed. Figure 9 depicts one of these cases while Atlas is walking across a rock-field.

8 Atlas-DRC

In August of 2012, we began to develop a version of Atlas to support the DARPA Robotics Challenge (DRC) program, an international contest designed to foster the development of advanced robotics technology. A barrier to entry for many robotics R&D groups is access to reliable, capable robotic hardware. To overcome this limitation, DARPA funded Boston Dynamics to build copies of Atlas-DRC. These copies would be supplied to teams selected to compete in the DRC based on their performance in a simulated robot contest. With less than 1 year to ship seven of these robots to DRC contestants, Atlas-DRC became a limited redesign of PETMAN. This redesign supported requirements for more capable arms, a sensor head, and an electrically tethered onboard HPU. A key to informing these redesigns was continued aggressive behavior development on Atlas meant to test the limits of the machine and demonstrate lifelike motion (Fig. 10).

Mass (kg)	152
Overall height (cm)	183
#Actuated DoFs (Total: Legs, Back, Arms, Neck)	28: 2×6, 3, 2×6 ^a , 1

^aw/o hands

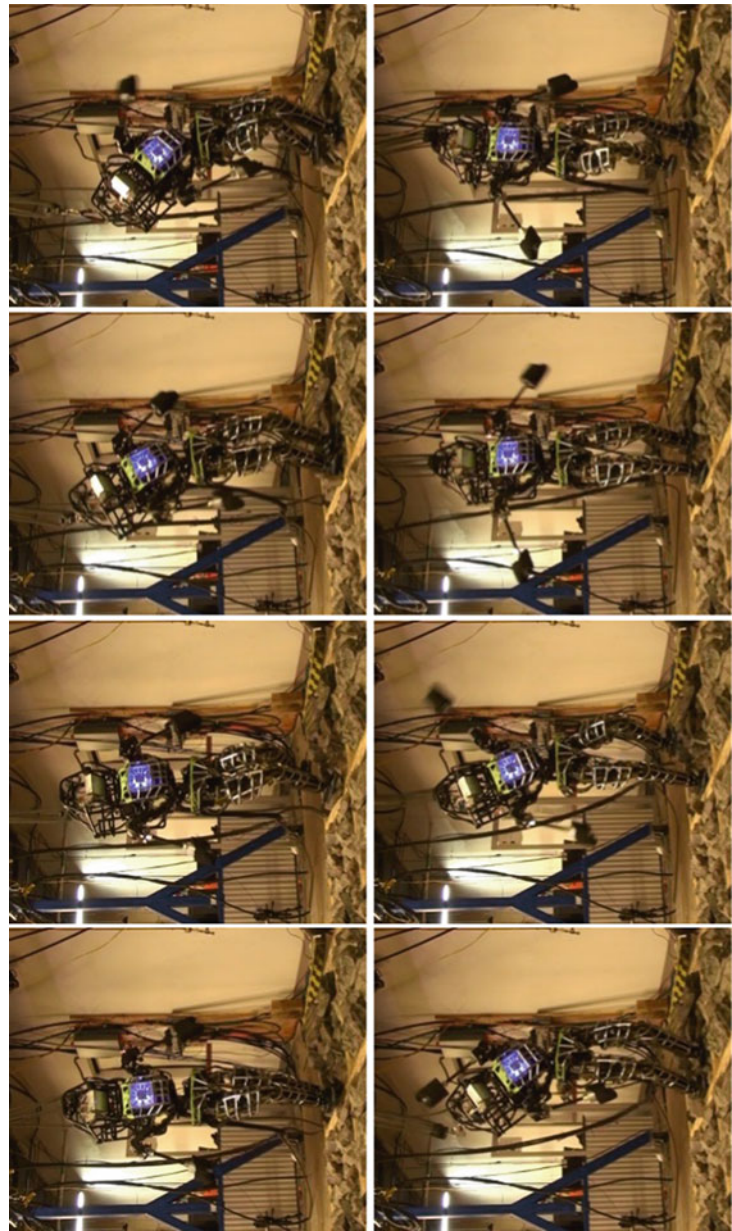


Fig. 9 Atlas' control could reflexively change the angular momentum of the robot to help it negotiate irregular terrain. Shown here are eight snapshots, at 200 ms intervals (*left-to-right* then *top-to-bottom*), of Atlas walking across a rock-field. The robot is disturbed when a rock rolls out from under the right foot causing that foot to slip to the robot's left. This elicits a fast, postural change, exhibited in the back then arms, which prevents the robot from subsequently falling to its right while it continues to walk across the rocks

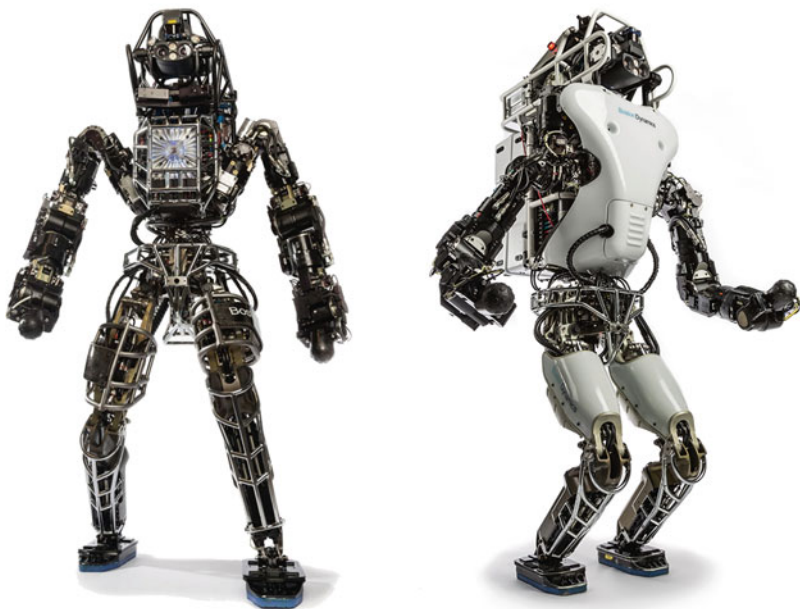


Fig. 10 Two variants of the Atlas were built especially for the DARPA Robotics Challenge. The DRC Preliminaries in December of 2013 used Atlas-DRC (*left*), operated with a tether that included electric power, cooling water, and communications. Atlas-Unplugged (*right*) was a significant redesign for the DRC Finals held in June 2015 in order to operate without a tether

The quality of PETMAN walking convinced us that we could reuse the lower body with minimal changes – small changes meant to reduce joint friction, backlash, and compliance. The sole exceptions were the feet. PETMAN had human-sized compliant feet that were good for walking fast but limited the ability of the robot to balance well while walking slowly or standing. For this reason, and as discussed above, we built a broad flat foot with integrated force sensing. Using this CoP sensing and control, we began to develop Capture Point-based stair climbing (Fig. 11).

Atlas-DRC also needed a perception head to collect and convey environmental data. We selected the Carnegie Robotics Multi-sense for these purposes. The Multi-sense was mounted on a single degree of freedom neck allowing the head to “nod” in the sagittal plane. Six DoF hydraulic arms were redesigned using a new rotational hydraulic actuator purpose built for this application. An onboard hydraulic pump generated power for 28 actuators.

Boston Dynamics provided an application programming interface (API) to the robot for the DRC teams, which ran controls software on the onboard computer. This API allowed access to sensor feedback, low-level joint controllers, and packaged high-level behaviors. Examples of these high-level behaviors include a standing

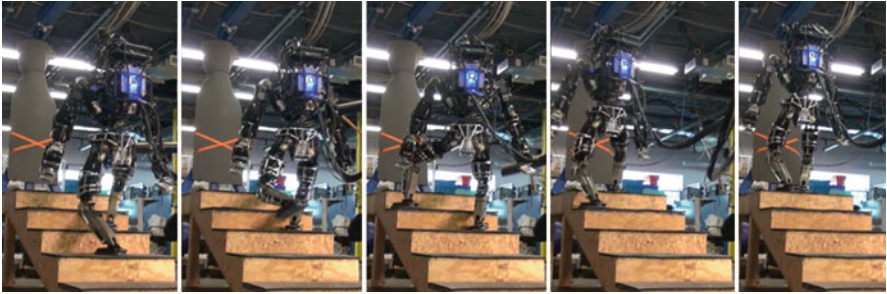


Fig. 11 Atlas-DRC climbing stairs using Capture Point control. The robot is pushed backwards and to its left in the second frame, causing that step to take longer and the left foot to step out further as it climbs. The mechanically timed aspect of Capture Point-based control is well suited for this task, allowing the robot to reliably step on each stair tread

behavior to balance the robot while using the upper body for manipulation tasks, a stepping behavior that allowed the robot to step in specified locations in irregular terrain, and a walking behavior for faster traversal of modest terrain. The DRC Trials tasks included driving a vehicle, opening various doors, turning a valve, cutting a hole in a wall with a power tool, negotiating a cinder block rubble pile, clearing debris, and climbing a ladder. DRC teams were free to use Boston Dynamics’ packaged behaviors to attempt parts of these tasks, or could develop their own behaviors. Teams using the Atlas-DRC placed 2nd, 4th, 6th, 7th, 8th, 10th, and 12th at the DRC Trials held in December of 2013 at the Homestead Miami Speedway in Florida.

9 Atlas-Unplugged

The next big step in Atlas’ evolution was to give the robot power autonomy and more computational resources – this in support of the DARPA DRC Finals. Using supervised autonomy, the DRC Finals required the contestant robots to complete a series of sequential tasks. These were driving and dismounting from a vehicle, opening a door, turning a valve, cutting a hole in a wall with a handheld power tool, negotiating a rubble pile, climbing stairs, and one surprise task. All tasks needed to be completed within 60 min. Should the robot fall down or otherwise become incapacitated, one human-assisted reset was allowed, albeit with a significant time penalty. Robot operators were sequestered away from the course and given a limited-bandwidth network connection to the robot.

Mass (kg)	182
Overall height (cm)	188
#Actuated DoFs (Total: Legs, Back, Arms, Neck)	30: 2×6, 3, 2×7 ^a , 1

^aw/o hands

Atlas-Unplugged was Boston Dynamics' first untethered biped robot. This required not only an onboard power plant and battery but a redesign of most of the major actuators to improve efficiency. Battery-powered operation allowed about 90 min of mixed-mission runtime, consisting of standing, walking, and manipulation. The upgrades to support power autonomy included adding an electrically driven fixed-displacement pump producing a variable supply-pressure system. Robot actions requiring low joint forces could use a lower pressure set-point. These changes made the robot much quieter, being nearly inaudible during low pressure operation. In contrast, Atlas-DRC required hearing-protection. Also, actuator sizes and joint linkages were redesigned for more efficient operation.

Atlas-Unplugged included a complete arm redesign. To improve the usable workspace for the arm, both the mounting setup and joint strength limits were adjusted, and a seventh DoF was added. The neck and three of the forearm DoFs were changed from hydraulic to electric actuation. Also, for lab testing and behavior development purposes, a battery emulator was created to simulate actual battery performance. Overall, the overhaul of Atlas-Unplugged required redesigns of about 70% of the Atlas-DRC parts.

There were various other computational upgrades to support supervised autonomy. The robot had a 1 Gbps WiFi connection to the outside world and a wireless E-stop. An onboard network, with a gigabit switch, linked three quad-core computers, dedicated to performer perception and control computations, to a dedicated Boston Dynamics controls computer which interfaced with the robot.

The DRC Finals were held in Pomona, California, in June of 2015. Throughout the program, and during the Finals, Boston Dynamics supported seven DRC teams who were using either Atlas-DRC or Atlas-Unplugged for the competition. Because of this support arrangement, Boston Dynamics could not compete in the DRC. Three of the top seven finishers in the final competition used Atlas-Unplugged. This included the second place team from the Institute for Human and Machine Cognition in Pensacola, Florida.

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